Traumatic Brain Injury

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Introduction

- ☑ Major public health problem
- ☑ Leading cause of death and disability for children and young adults
- ☑ Can still be significant without loss of consciousness
- ☑ Can be divided into traumatic brain and traumatic spinal cord injury
- ☑ Mode of injury can be by blunt, sharp, and compression force

Clinical Features

History

- ☑ *Injury*:
  - Timing
  - Type of accident
  - Degree of force acting on the head
  - Position of the victim when found at the scene
  - Victim's immediate state of consciousness

  Severe mechanism includes:
  - Motor vehicle accident with patient ejection or rollover, death of another passenger
  - Pedestrian or cyclist struck by motor vehicle
  - Fall >1.5 m (>2 years)
  - Head struck by a high-impact object
- ☑ *Clinical high-impact associated symptoms*:
  - Loss or impairment of consciousness and duration
  - Abnormal behaviors, including agitation, confusion, and drowsiness
  - Headache

Clinical Features

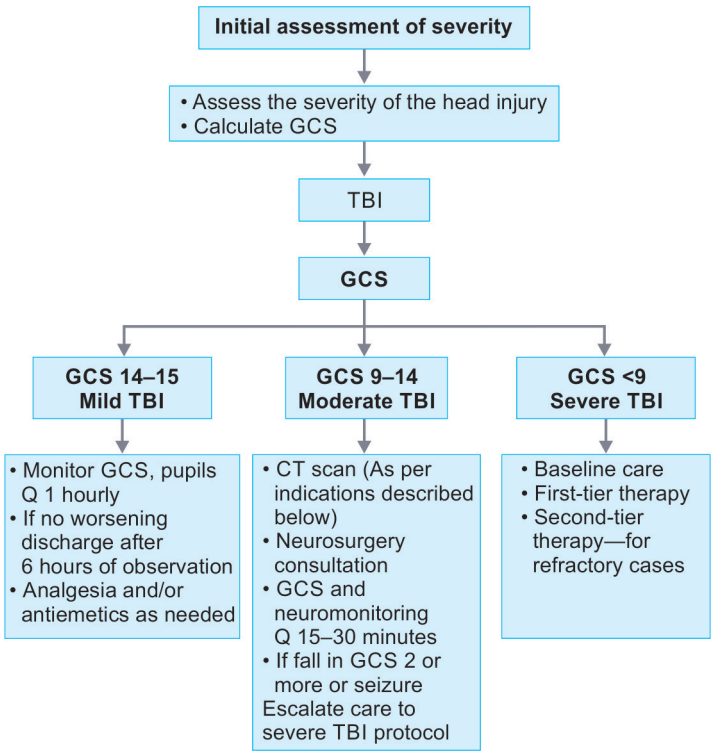
History

- Nausea and vomiting
- Other injuries sustained
- Amnesia

  Examination:
  - Unilateral or bilateral pupillary dilatation
  - Drop of >2 points in Glasgow Coma Scale (GCS)
  - Development of focal neurological signs
  - Abnormal posturing
  - Irregular respirations, hypertension, and bradycardia
- Cushing reflex-late sign
- Feel specifically for palpable skull fractures
- Look for signs of the fractured base of the skull [hemotympanum, cerebrospinal fluid (CSF) otorrhea, or rhinorrhea, periorbital bruising (raccoon eyes), and bruising over the mastoid area (Battle sign)

Assessment and Classification

Flowchart 1: Assessment for traumatic brain injury.



(GSC: Glasgow Coma Scale; TBI: traumatic brain injury)

Modified Glasgow Coma Scale

≥4 Years

Response	Score
Eye opening	
Spontaneous	4
To speech	3
To pain	2
None	1
Best verbal response	
Oriented	5
Confused	4
Inappropriate words	3
Incomprehensible sounds	2
No response	1
Best motor response	
Obeys commands	6
Localizes pain	5
Withdraws to pain	4
Abnormal flexion	3
Abnormal extension	2
None	1

<4 Years

Response	Score
Eye opening	
Spontaneously	4
To verbal stimuli	3
To pain	2
No response to pain	1
Best verbal response	
Alert and babbles	5
Less than usual words	4
Cries only to pain	3
Moans to pain	2
No response to pain	1
Best motor response	
Spontaneous or obeys verbal commands	6
Localizes to pain or withdraws to touch	5
Withdraws from pain	4
Abnormal flexion to pain	3
Abnormal extension to pain	2
No response	1

Indication for Neuroimaging in Blunt Head Trauma

PECARN criteria for CT scan in blunt head trauma.

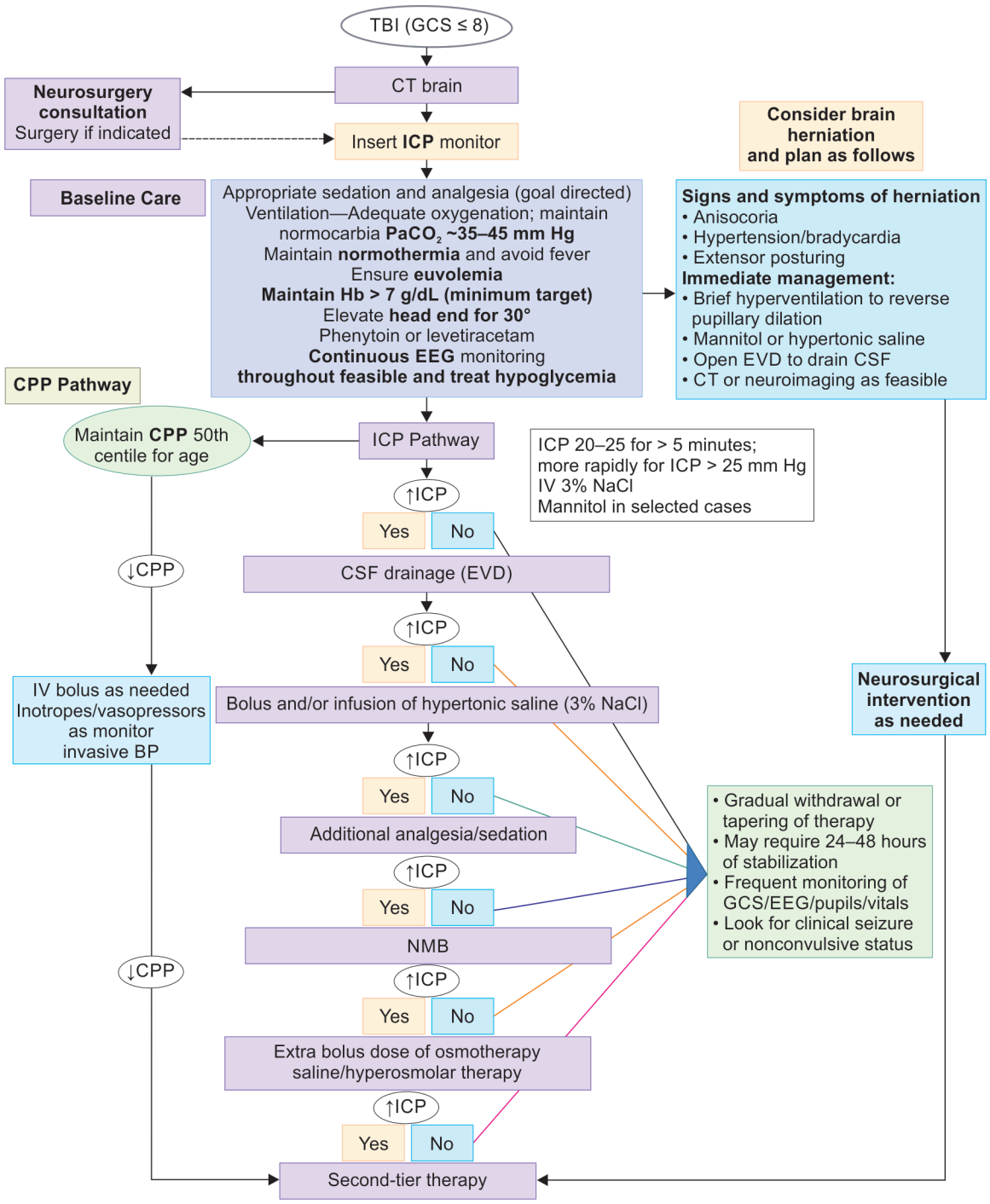
Clinical criteria	***Recommendation for CT head***
Children <2 years	
GCS ≤14 or altered mental status or palpable skull fracture	CT
GCS > 14, no altered mental status, no palpable skull fracture, occipital, parietal or temporal scalp hematoma, or loss of consciousness ≥ 5 seconds, or severe mechanism of injury or not acting normally	Observation versus CT depending on physician experience, multiple versus isolated findings, worsening symptoms or signs during observation, age <3 months, parental preference
Children >2 years	
GCS ≤14 or altered mental status or signs of basilar skull fracture	CT
☑ GCS >14, no altered mental status, no signs of basilar skull fracture ☑ History of loss of consciousness or vomiting or severe headache or severe mechanism of injury	Observation versus CT depending on physician experience, multiple versus isolated findings, worsening symptoms or signs during observation, age < 3 months, parental preference

Management of Severe Traumatic Brain Injury

Baseline Care

- ☑ *Neutral head positioning*: Keep the head elevated by 30° and in a midline position to optimize venous drainage from the brain without hampering cerebral perfusion.
- ☑ *Temperature management*:
  - Core body temperature is to be kept in the normothermic range.
  - Care must be taken to prevent fever.
  - Antipyretics (paracetamol—10–15 mg/kg body weight) should be given as soon as possible in a febrile patient.
- ☑ *Analgesia and sedation*:
  - The optimum dosage of opioid analgesics and benzodiazepines for sedation to be used to keep the patient calm.
  - The combination of fentanyl and midazolam is suitable in the majority of cases.
  - Judicious usages of the above drugs have to be done to avoid drug-induced hypotension.
- ☑ *Fluid therapy to maintain appropriate intravascular volume*:
  - *Initial fluid* is to be started as 75% of maintenance
  - *Fluid of choice*: Normal saline (NS) for the older child and 5% dextrose + NS (DNS) for infants and younger patients

Flowchart 2: Management of severe TBI.



(BP: blood pressure; CPP: cerebral perfusion pressure; CSF: cerebrospinal fluid; EEG: electroencephalography; EVD: external ventricular drain; GCS: Glasgow Coma Scale; Hb: hemoglobin; ICP: intracranial pressure; IV: intravenous; TBI: Traumatic brain injury;

Baseline Care

Goals of fluid therapy:

- *Maintain euvolemia:* Target blood pressure (BP) in the normotensive range (around 50th centile for age)
- *Target urine output:* Around 1 mL/kg/hour
- *Central venous pressure monitoring (CVP):* If CVP monitoring facilities are available keep target CVP between 8 and 12 mm Hg
- *Normoglycemia:* Frequent blood sugar monitoring [random blood sugar (RBS)] to avoid hypo/hyperglycemia
- *Eunatremia:* The rate and type of fluid should be carefully titrated to maintain serum sodium (Na^+) between 140 and 150 mEq/L
- During osmotherapy with 3% NaCl, serum sodium up to 160 mEq/L can be allowed
- *Feeds/nutrition:* Feeds are to be started as early as feasible, ideally within 72 hours and preferably by the nasogastric route

☑ *Prevention and/or treatment of anemia:* To maintain appropriate delivery of oxygen (DO_2), it is recommended to maintain a hemoglobin level of at least 7 g/dL.

☑ *Mechanical ventilation:*

- *Any mode* (pressure control or volume control) of ventilation can be used during the management of severe traumatic brain injury (TBI) patients.
- *Maintain normocarbia*:
  - Mechanical ventilator settings should be regulated to target partial pressure of carbon dioxide in arterial blood ($PaCO_2$) at the range of 35–40 mm Hg.
  - To achieve the goal of $PaCO_2$, the minute ventilation (ventilator rate × tidal volume) should be optimized depending on the lung condition
- *Target normoxia:* Optimize ventilation strategy to maintain pulse oximetry oxygen hemoglobin saturations (SpO_2) between 92 and 99% and if arterial blood gas is available then the target PaO_2 should be kept between 75 and 100 mm Hg.

☑ *Prevention and/or treatment of seizure:*

- *Antiepileptic drug therapy:* In high-risk patients, phenytoin or fosphenytoin or levetiracetam can be added prophylactically.
- Levetiracetam or phenytoin can be used as first-line therapy for clinical seizures.
- *Electroencephalography (EEG):*
  - Continuous EEG (cEEG) monitoring is needed in patients on neuromuscular drug therapy and in patients with recent episodes of seizure.
  - The cEEG monitoring also helps in the detection of nonconvulsive status.

☑ *Management of coagulopathy and thrombocytopenia*:

- Routine use of platelets and fresh frozen plasma (FFP) is not recommended to correct the altered laboratory parameters.
- If intracranial pressure (ICP) monitoring catheter insertion is planned then FFP transfusion is required if the international normalized ratio (INR) is >1.5 and platelet transfusion is indicated to maintain platelet counts above 50,000–100,000/mm^3.

First-tier Therapies

- ☑ ICP pathway (needs invasive ICP monitoring device)
- ☑ Cerebral perfusion pressure (CPP) pathway
- ☑ Brain tissue oxygen ($PbrO_2$) pathway—needs invasive monitoring and is rarely available in resource-limited settings.

Intracranial Pressure Pathway

- ☑ *ICP target:* If the ICP catheter is in place then the target ICP should be kept below 20 mm Hg.
  If ICP remains >20 mm Hg for 5 minutes then immediate additional interventions have to be initiated like a bolus dose of sedative or analgesics and osmotherapy.
- ☑ *Osmotherapy:* Hypertonic saline (3% NaCl) at a bolus dose of 1–3 mL/kg (maximum 25 mL) over 10 minutes followed by infusion at a rate of 0.5–1 mL/kg/hour to be started.
- ☑ *During hypertonic saline therapy:* The target maximum sodium should be <160 mEq/L and osmolality should be <360 mOsm/kg.
- ☑ Though hypertonic saline is the preferable osmotherapy agent, IV mannitol (2.5 mL/kg) over 20 minutes can be used as rescue therapy if the patient is normotensive, has no kidney injury and serum osmolality is < 20 mOsm/kg.
- ☑ *Refractory raised ICP/failed ICP pathway:* If ICP remained elevated despite osmotherapy, additional sedation and/or analgesics can be given as bolus doses along with neuromuscular blocking agents MBs).

Elevated ICP despite the optimum dosage of medications mandates initiation of second-tier therapy.

Cerebral Perfusion Pressure Pathway

- ☑ CPP = mean arterial pressure (MAP) – ICP
- ☑ *In absence of ICP monitoring*: Keep systolic blood pressure (SBP) or MAP >50th centile for age by titrating fluids (if needed NS fluid bolus).
- ☑ If BP is not reaching till 50th centile by fluids alone, judicious use of vasopressors (like norepinephrine or adrenaline) has to be done.

Second-tier Therapy

- ☑ Neurosurgical interventions and/or any of the following (A/B/C) to be used alone or in combinations.
- ☑ Neurosurgery consultations to be taken in refractory raised ICP following TBI for the following probable options:
  - External ventricular drainage (EVD) placement and CSF drainage
  - Removal or evacuation of hematoma
  - Decompressive craniectomy

Management of Severe Traumatic Brain Injury

Second-tier Therapy

- ☑ *Barbiturate infusion*:
  - *Commonly used drug*—Thiopentone infusion at 2–4 mg/kg/hour
  - Hemodynamic monitoring is needed to prevent hypotension during barbiturate infusion
  - Gradual tapering of infusion has to be done if ICP remained <20 mm Hg for 24 hours
- ☑ *Moderate hypothermia:* A target temperature of 33–35°C can be used as a late option for refractory cases
- ☑ *Hyperventilation:* A brief period of hyperventilation can be tried to keep a target of a $PaCO_2$ between 28 and 34 mm as rescue therapy.

Pearls on Endotracheal Intubation in Severe Pediatric TBI Indications

- ☑ *Indications:*
  - Absent gag reflex
  - For airway protection and mechanical ventilation
  - Poor breathing efforts
- ☑ *Precautions:*
  - Ensure cervical spine immobilization during intubation
  - Midazolam + fentanyl + IV lignocaine + muscle relaxant ± thiopentone can be used
  - Keep IV NS (0.9% NaCl) for bolus therapy and inotropic infusion ready to prevent hypotension in the periintubation period

Further Reading

- ☑ Gambacorta A, Moro M, Curatola A, Brancato F, Covino M, Chiaretti A, et al. PECARN Rule in diagnostic process of pediatric patients with minor head trauma in emergency department. Eur J Pediatr. 2022;181(5):2147-54.
- ☑ Kochanek PM, Tasker RC, Bell MJ, Adelson PD, Carney N, Vavilala MS, et al. Management of Pediatric Severe Traumatic Brain Injury: 2019 Consensus and Guidelines—Based Algorithm for First and Second Tier Therapies. Pediatr Crit Care Med. 2019;20(3):269-79.
- ☑ Kochanek PM, Tasker RC, Carney N, Totten AM, Adelson PD, Selden NR, et al. Guidelines for the management of pediatric severe traumatic brain injury, third edition: Update of the brain Trauma Foundation Guidelines. Pediatr Crit Care Med. 2019;20(3S Suppl 1):S1-S82.
- ☑ Lui A, Kumar KK, Grant GA. Management of Severe Traumatic Brain Injury in Pediatric Patients. Front Toxicol. 2022;4:910972.

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STANDARD TREATMENT GUIDELINES 2022

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